

TOI-2046b

R-2358

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI2046_220913 · created 2026-08-19T23:01:42+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2459835.82 +/- 0.14 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.154 +/- 0.047
Transit depth (Rp/Rs)^2	2.37 +/- 1.44 [%]
Orbital Inclination (inc)	87.86 +/- 2.87
Airmass coefficient 1 (a1)	0.919 +/- 0.093
Airmass coefficient 2 (a2)	0.07 +/- 0.26
Scatter in the residuals of the lightcurve fit is	10.2 %
Best Comparison Star	#2 - [250, 264]
Optimal Aperture	1.51
Optimal Annulus	15.1
Transit Duration (day)	0.034 +/- 0.042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.154 ± 0.047 vs 0.1213 (deviation 0.7σ)
Duration fit vs published	0.034 d vs 0.1033 d (deviation 1.65σ)
Mid-time O–C	-165.7 min
Depth SNR	3.3
Residual scatter	10.2 %

Light curve

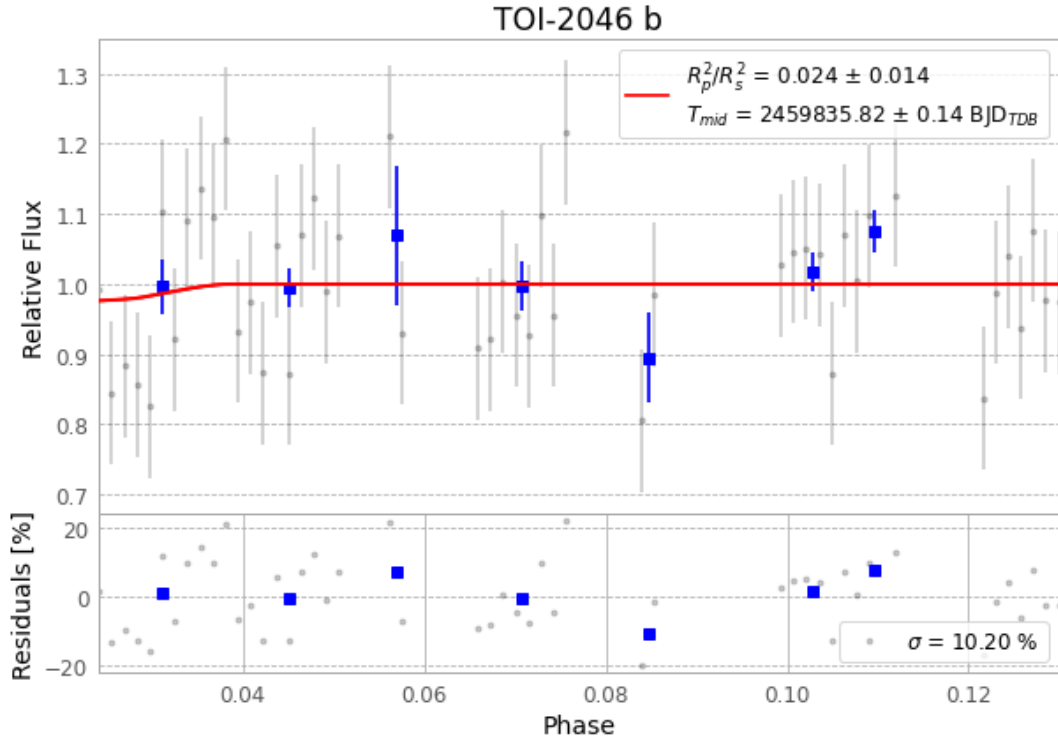


Figure 1 — FinalLightCurve_TOI-2046 b_2022-09-13.png (SHA-256 d0d6ab991924c949...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [275, 415], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 758e2b1615be705a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

77 FITS frames (51 MB), TOI2046220913082413.FITS → TOI2046220913121213.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI2046-220913.log (f025e21428f8...), FinalLightCurve_TOI-2046 b_2022-09-13.png (d0d6ab991924...), FinalLightCurve_TOI-2046 b_2022-09-13.csv (35c6cc9ca498...)

Compute resources

Wall time 600.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-19 23:01Z.